

B.Tech. (Civil Engineering)
End Semester Examination 2024
VII Semester
Construction Management and Planning

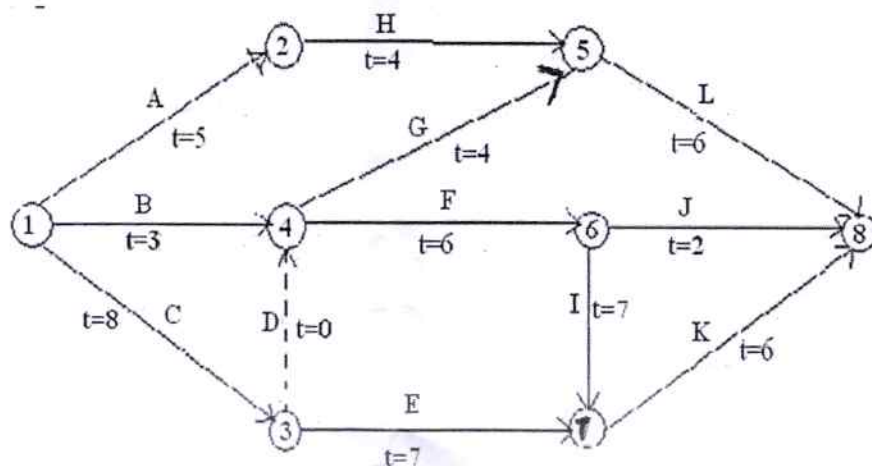
Time: Three Hours

MM: 100

Note:

- (i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.

- Q1. (2X10=20 Marks)
a. Describe the importance of financial evaluation of projects and project planning? CO1
b. Explain net present value and benefit cost ratio? CO1
c. Explain main causes of project failure in construction industries? CO1
- Q2. (2X10=20 Marks)
a. Describe the objectives, project development process in construction management? CO2
b. Define project planning and describe the steps involved in project planning? CO2
c. Describe the objectives of construction project management and explain the importance of construction project management in construction industries? CO2
- Q3. (2X10=20 Marks)
a. Explain different elements present in PERT network and explain with an example CO3
b. Explain the differences between Critical Path Method and PERT technique CO3
c. The network of a certain project is shown in Figure.1 with the estimated durations of various activities. Determine the following: (i) Earliest event time and latest event time (ii) Earliest and latest start and finish times of each activity. (iii) Total and free floats for each activity. (iv) Critical path for the network



CO3

- Q4. (2X10=20 Marks)
a. Explain a typical cost – duration curve and show on the optimum duration and minimum project Cost. CO4
b. Discuss the importance of Resource Allocation in construction industry CO4

- c. The direct cost of the project is Rs 2000 per week. Determine the optimum duration of the project and the corresponding minimum cost. Draw the least cost network.

Activity	Normal duration (weeks)	Normal cost (Rs)	Crash duration (weeks)	Crash cost (Rs)
1-2	4	4000	2	12000
2-3	5	3000	2	7500
2-4	7	3600	5	6000
3-4	4	5000	2	10000

CO4

Q5. (2X10=20 Marks)

- Explain contract management and describe the various elements of tender operation? CO5
- Explain Contract negotiation & award of work and their importance in construction management? CO5
- Explain safety guidelines in construction activities CO5